H MANJUNATHA

Bengaluru, Karnataka | +91 7353403062 | manjunathah735@gmail.com | LinkedIn | github

CAREER OBJECTIVE

B.E. graduate in Electrical and Electronics Engineering with Embedded Systems training from Emertxe Information Technologies. Skilled in Embedded C, Advanced C, Linux Programming, PIC18F4580, communication protocols, and firmware development. Seeking an Embedded Software Engineer role to contribute to real-time embedded product development.

TECHNICAL SKILLS

- Programming: C, Embedded C, Basic C++
- Microcontrollers: PIC18F4580
- Protocols: UART, SPI, I2C, CAN
- Embedded: GPIO, Timers, Interrupts, ADC, PWM, Bitwise Programming
- Tools: MPLAB X IDE, XC8, GCC

PROJECTS

CAN-Based Automotive Dashboard — Multi-Controller Vehicle Monitoring System

- Architected a 3-controller CAN bus network to display real-time RPM, speed, gear position, and indicator status on a car dashboard.
- Implemented interrupt-driven message handling to keep multi-node data synchronized with no observed message collision under test conditions.

Image Steganography Tool — LSB Encoding/Decoding

- Developed an encoder/decoder pair that hides secret file data inside BMP pixel data using LSB substitution with zero visible image distortion.
- Preserved magic-string and metadata integrity across the encode/decode cycle to guarantee lossless data recovery.

Address Book Management System

- Developed a menu-driven CLI system to add, delete, modify, search, and sort contacts by name, phone, or email with persistent file storage.
- Implemented input validation to catch duplicate contacts and invalid data formats across sessions.

ACADEMIC PROJECT

Automated Paralysis Patient Health Care System

- Developed an IoT-based healthcare system using Arduino Uno, accelerometer, DHT11, NodeMCU, 16×2 LCD, and Blynk to monitor patient movement and environmental conditions.
- Implemented gesture-based communication to convert patient hand movements into messages such as *I Need Food, I* Need Water, I Need Washroom, *and* Emergency, displayed on the LCD and transmitted through Blynk for remote monitoring.

EDUCATION

Bachelor of Engineering (E&E) 2025

Jain Institute of Technology, Davangere (VTU) — 82.29%

Diploma (E&E) 2021

STJ Polytechnic, Harapanahalli — 65%

Secondary(X), State Board 2018

LHS High school, Hampasagara — 72%

CERTIFICATION

• Emertxe Certified Embedded Professional — Spot Award, Jan 2026
• Certified in Embedded C, Advanced C, Data Structures & Algorithms, C++ Programming, and Linux